

Ali Shafiee

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[Codeforces](#) | [LinkedIn](#) | [IOI Profile](#) | [Google Scholar](#)

IOI silver medalist and ICPC World Finalist. MSc researcher in probability, stochastic control, and algorithmic game theory.

EDUCATION

Institute of Science and Technology Austria (ISTA)

Vienna, Austria

MSc in Computer Science, grade average 1.0 (Austrian scale; 1.0 highest), advisor Prof. Krishnendu Chatterjee Sep. 2025 – present

Sharif University of Technology

Tehran, Iran

BSc in Computer Engineering, GPA 3.68/4 (17.03/20)

Sep. 2018 – Aug. 2023

RESEARCH AND PUBLICATIONS

Complexity and algorithms for **sequential decision-making under uncertainty**: Markov decision processes with partial observation and with adversarial model uncertainty, plus **algorithmic game theory**. Author lists are alphabetical, per convention in the field.

- [Randomise Alone, Reach as a Team](#). L. Brice, T. A. Henzinger, A. Montaseri, **A. Shafiee**, K. S. Thejaswini. *CAV 2026*.
- [Qualitative Analysis of \$\omega\$ -Regular Objectives on Robust MDPs](#). A. Asadi, K. Chatterjee, E. Goharshady, M. Karrabi, **A. Shafiee**. *AAAI 2026*.
- [Limit-Sure Reachability for Small-Memory Policies in POMDPs is NP-Complete](#). A. Asadi, K. Chatterjee, R. Saona, **A. Shafiee**. *UAI 2025*.
- [Quantitative Analysis of \$\omega\$ -Regular Robust MDPs](#). A. Asadi, K. Chatterjee, E. K. Goharshady, M. Karrabi, A. Montaseri, **A. Shafiee**. *AAAI 2027*, under review.
- [On the Complexity of Discounted Robust MDPs with \$L_p\$ Uncertainty Sets](#). A. Asadi, K. Chatterjee, A. Montaseri, **A. Shafiee**. *NeurIPS 2026*, under review.
- Mechanism Design for Multi-Resource Nakamoto Consensus. N. Babashah, M. A. Baig, K. Pietrzak, **A. Shafiee**. *WINE 2026*, under review.

COMPETITIVE PROGRAMMING

- **Codeforces: top 50 worldwide in 2019**; International Grandmaster, peak rating 2800 (handle ToTLeS)
- **Silver Medal, International Olympiad in Informatics (IOI)**, 2017: 76th of 300+ contestants from 80 countries
- **33rd Place, ACM-ICPC World Finals**, Moscow 2021: among 10,000+ teams from 110 countries
- **Gold Medal, ACM-ICPC West Asia Regional Contest**: 1st place 2019, 2nd 2018, 3rd 2020; 100+ teams each year
- **Silver Medal, Asia-Pacific Informatics Olympiad (APIO)**, 2018 and 2017: 20+ participating countries
- **Gold Medal, Iranian National Olympiad in Informatics**, 2016: top 8 of 10,000+ students nationally
- **Bronze Medal, Iranian Mathematical Olympiad**, 2017: top 80 of 20,000+ students nationally

INDUSTRY EXPERIENCE

Quera, Iran's largest competitive-programming and technical-education platform

Tehran, Iran

Contest Manager, Content Manager, Director of CS Education, Business Line Manager

Mar. 2019 – Aug. 2025

- Owned a business line end to end (Oct. 2024 – Aug. 2025); previously led a 40-person division spanning six technical teams (back-end, front-end, data, blockchain, algorithms, fundamentals) and set the education product roadmap.
- Designed and shipped 12+ bootcamps and courses in C++, Python, SQL, Django, Linux/Git, data analysis, machine learning, and deep learning.
- Ran 20+ large-scale programming contests, including sponsored hiring hackathons with corporate partners.

TEACHING & SERVICE

- **Scientific Committee**, ACM-ICPC West Asia Regional Contest (2024 – present); **Host Scientific Committee**, [IOI 2019](#), Baku
- **Problem-setter** (2017 – present): Iranian National Olympiad in Informatics, Codecup, and Quera contests
- **Teaching Assistant**, Sharif University: Design of Algorithms, Data Structures & Algorithms, Fundamentals of Programming
- **Olympiad coach** (2017 – 2023): trained 50+ students for the Iranian National Olympiad in Informatics and the IOI

SKILLS

Programming: C++, Python, Rust · **Numerical & ML**: NumPy, SciPy, pandas, Matplotlib, scikit-learn, PyTorch, JAX

Tools: Git, Linux/Bash, SQL, Docker, Jupyter, L^AT_EX · **Mathematics**: probability, stochastic processes, Markov decision processes, optimization, game theory

Languages: Persian (native), English (TOEFL 110/120)